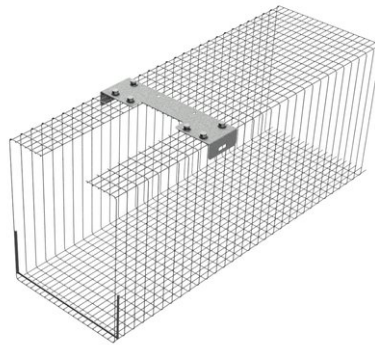




OcuTrap R2

Installation & User Manual



Please read this manual fully before assembling or operating your OcuTrap R2.



Online docs & video
guides
docs.ocutrap.com

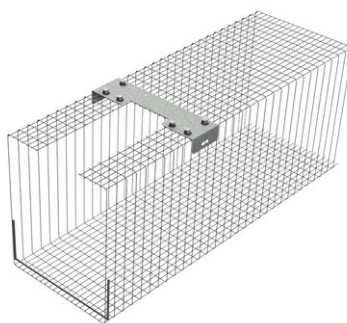
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Unboxing and Initial Inspection

Carefully unpack your OcuTrap R2 and check that everything is there:

- **Cage**, with the door and motor already installed
- **Parts box**: battery, charger, POD, quick start card



OcuTrap cage

Before you continue

Examine each item for any visible damage. If anything is missing or appears damaged, halt installation and contact OcuTrap Support at support@ocutrap.com with your trap ID.

About the photos

Photos of the R2 cage are coming. The POD, battery, buttons, LEDs, and app are identical on the R2 and the R1, so those photos and screenshots apply to both models.

Hardware Setup

The OcuTrap R2 ships with the door and the motor already installed. Setup takes about 10 minutes and needs no tools.

1. Charge the battery
2. Cut the motor zip tie
3. Mount the POD on the back
4. Insert the battery and power on
5. App setup

Charge the Battery

Fully charge the blue battery **first**, using the charger found in the small white box. The charger light is **red** while charging and **green** when full. The R2 ships with the **5,200 mAh (56 Wh)** pack and its **1 A** charger. A full charge takes about **5 to 6 hours**, so start it now and set up the rest of the trap while it charges.



If anything appears damaged, contact support@ocutrap.com for assistance.

Cut the Motor Zip Tie

The motor is zip-tied to the cage for shipping. Remove the tie before you power the trap on.

1. Cut the zip tie with snips or scissors.
2. Pull the cut tie free and discard it.

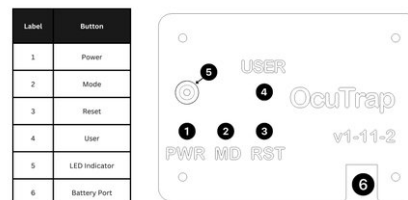
■ Caution: Moving door

- Keep hands and fingers clear of the door path. The door can swing once the tie is cut.
- **Cut the zip tie only. Do not cut any wire.**

Mount the POD on the Back

The POD holds the camera, the distance sensor, and the electronics. It slides onto rails on the back of the cage.

1. Slide the POD down the rails on the back of the trap until it is in place.
2. Attach the clip at the top to keep the POD in place.
3. Attach the motor's wire to the POD using the locking screw connector, ensuring a secure connection.
4. Use the top latch to secure the POD in place.



Insert the Battery and Power On

1. Open the POD latch.
2. Place the charged battery in the bracket at the bottom of the POD.
3. Plug the battery into the POD using the yellow connectors. Push straight in until fully seated.
4. Tuck the cables along the edge of the enclosure, clear of the door closing points, so the wires do not get squished.
5. Close the latch fully to maintain waterproofness.
6. Press the **Power** button to turn the trap on.

Watch the status LED. **Breathing cyan** means the trap is connected to the cloud and ready. Finding cellular can take up to 10 minutes on first power-up.

App Setup

Step 1: Create an account

Go to **base.ocutrap.com** (or scan the QR code on the right) and create an account.



Create an account

Step 2: Download the mobile app



iOS download



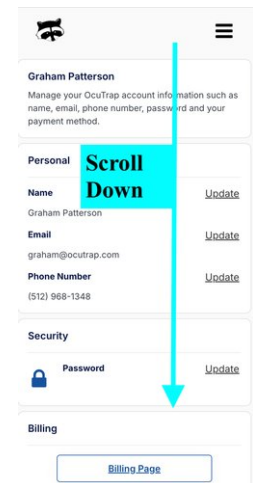
Android download

Step 3: Activate the trap through the app

1. Create an account on the app.
2. Locate the POD for the OcuTrap you wish to register, open it, and find the serial number on the top of the device.
3. In the app, go to **Account** (top right of the dashboard), then tap **Add Trap** at the bottom of the page.
4. Enter the serial number when prompted.
5. Follow the in-app prompts to enable your subscription.
6. Once complete, the new trap will appear in your dashboard.



Account page — scroll down



Tap "Add Trap"

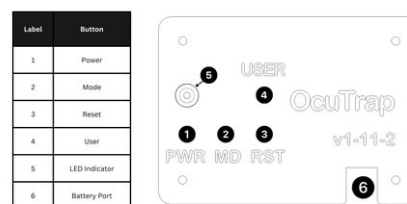
Initial hardware checkout

1. **Power on:** connect the fully charged battery.
2. **Connectivity check:** ensure the trap is within network range. Check the indicator light for a stable connection (breathing cyan). It may take up to 10 minutes to connect to cellular networks.
3. **Open the control panel:** open the OcuTrap app.
4. **Engage the trap:** select **Arm** in the app to initiate trap readiness. Ensure cameras, sensors, and mechanisms respond correctly.

Operating the POD

Buttons on the POD

Label	Button
1	Power
2	Mode
3	Reset
4	User
5	LED Indicator
6	Battery Port



Arming / Unarming

Use the **User Button** and **Power Button** together to toggle between **Armed** and **Unarmed** states.

Action	Steps	Indicator
Arm the device	1. Press the User Button. 2. Immediately press the Power Button. 3. Ensure the door is set to open before arming.	Flashing yellow (arming)
Unarm the device	1. Press the User Button. 2. Immediately press the Power Button again.	Flashing white (unarmed)

Reset

Restarts the device **without changing its configuration**.

Action	Steps	Indicator
Reset the device	1. Press the Reset Button once. 2. The device will reboot and retain its current state.	No LED indicator

Door Functionality

Action	Steps	Indicator	Description
Open door	1. Press the User Button. 2. Release, then press and hold for 5 seconds.	Blue flashes	Door opens
Close door	1. Press the User Button. 2. Release, then press and hold for 5 seconds.	Green flashes	Door closes
Arm / Unarm	Follow the Arming / Unarming steps above.	See above	Toggles Armed / Unarmed

Device Status Indicators

Mode	Color / Pattern	Description
Unarmed & Open	Solid Blue	The device is unarmed , and the door is open .
Unarmed & Closed	Solid Green	The device is unarmed , and the door is closed .
Armed Mode	Solid Yellow	The device is armed .
Armed & Captured	Solid Magenta	The device is armed , and a capture event has occurred.

Shut-down Operation

Steps	Indicator
Power off: hold down the PWR button for 3 seconds	Blinking Red
Shutting down (takes about 10 seconds)	Solid Red and other lights
Fully powered off	No LED

Weather

OcuTrap is designed to withstand various outdoor conditions, including rain, wind, and moderate environmental challenges. While the enclosure is weatherproof, internal electronics should never be submerged in water. The trap is built for durability, but users should take precautions during extreme weather to prevent potential damage. For optimal performance, OcuTrap operates best between 0°C (32°F) and 40°C (104°F). Prolonged exposure to temperatures outside this range may affect performance and battery longevity. The lithium-ion battery should be charged at room temperature to maintain efficiency and lifespan.

Extreme heat and cold can reduce performance and battery life, while freezing conditions may cause the door mechanism to become stuck and the motor to not operate properly. To prevent icing, avoid operating the trap in environments where freezing is likely. Although the enclosure is weatherproof, heavy rain or flooding could compromise internal electronics, so relocating or protecting the trap is advised. To maximize battery longevity, follow the recommended temperature guidelines and avoid prolonged exposure to extreme conditions. No additional maintenance is required.

Routine Maintenance

Run this checklist after every capture, and after harsh weather.

1. Cleaning

Item	Action
Camera and sensor lens	Wipe with a soft, lint-free cloth until clear.
Exterior	Wipe with a damp cloth.
Interior parts (door, motor area)	Remove any debris, wipe if needed.

2. Battery

Task	Action
Charge	Fully charge before use, top off as needed.
Terminals	Check for corrosion or wear, wipe dry if dirty.

3. Quick Function Check

System	Action
Door	Test open and close, fix any sticking.
Motor and sensors	Confirm motor runs and sensors respond.

4. Weatherproofing

Area	Action
Seals, enclosures	Check for cracks or looseness.
After bad weather	Look for moisture or debris inside, dry out if needed.

5. Log

Keep a simple note of cleanings, checks, and repairs.

Safety

Power off the trap and unplug the battery before maintenance.

LED Indicator Reference

Status	Color / Pattern	Description
Connected	Solid Cyan	Device is successfully connected to the cloud.
Connecting	Blinking Cyan	Device is establishing cloud connection.
Firmware Update	Blinking Magenta	Device is performing an OTA update.
Internet Search	Blinking Green	Device is searching for internet connection.
SOS Alert	Flashing Red (•• ———)	Device requires immediate attention.
Low Power	Blinking Red	Battery level too low for operation.
Error State	Solid Red	System error detected.
No Power	No Light	Device is either powered off, in ultra-low power mode, in armed low power mode, or requires power source check/reset.

Battery Safety Precautions

The OcuTrap uses sealed lithium-ion batteries. **Dispose of used batteries according to all applicable federal, state, and local regulations.**

General Safety Warnings

- Do not expose the battery to fire, water, or excessive heat.
- Avoid short-circuiting the terminals.
- Use only the provided charger to prevent overcharging or damage.
- Handle the battery carefully to prevent punctures or impacts.
- Keep out of reach of children.

Base Battery

The KBT 12V Rechargeable Li-ion Battery is a versatile power source for 12V devices. Designed for reliability and long-term use, it is equipped with essential safety features and a durable build.

Specifications

- Voltage: 12V
- Battery type: Lithium-ion
- Charging cycles: over 800 cycles

Charging Instructions

1. Use the provided 12V charger for optimal performance.
2. Connect the barrel connector securely to the charger's barrel connector.
3. Charge the battery in a cool, dry place.
4. Avoid leaving the battery connected to the charger once fully charged.

Battery Installation in the POD

1. Open the POD latch.
2. Place the battery in the bracket at the bottom of the POD.
3. Plug the battery into the POD using the yellow connectors. Push straight in until fully seated.
4. Make sure the cables are tucked out of the way of the door closing points, along the edge of the enclosure, so the wires do not get squished.
5. Close the latch fully to maintain waterproofness.

Maintenance & Storage

- Store the battery in a cool, dry place away from direct sunlight or heat sources.
- For long-term storage, keep the battery partially charged (around 40–60%) to preserve its health.
- Clean the terminals periodically to maintain proper conductivity.
- Avoid exposing the battery to moisture or corrosive materials.

Troubleshooting FAQs

Q: What should I do if the battery doesn't charge?

A: Verify the charger is functioning properly and ensure secure connections to the terminals.

Q: Why is the battery overheating?

A: Stop usage immediately and check for device compatibility or possible overcharging.

Q: How can I extend the battery life?

A: Avoid fully depleting the battery before recharging and store it properly when not in use.

Cautions

- Do not attempt to open or modify the battery.
- Avoid charging the battery in extreme temperatures ($<0^{\circ}\text{C}$ or $>40^{\circ}\text{C}$).
- Do not connect the battery to devices exceeding its voltage or capacity limits.
- Dispose of the battery properly according to local regulations when it reaches the end of its lifespan.

Safety Warnings — Moving Parts & Animal Handling

■ Caution: Risk of Finger Injury

The OcuTrap's moving parts, including the motorized door and spring mechanism, can cause serious injury if fingers are placed in or near the trap's assembly while in operation.

- Keep hands and fingers clear of the trap door and motorized components at all times.
- Ensure the trap is powered off before performing maintenance or adjustments.
- Always use the provided assembly tools to adjust or interact with components.
- Do not allow children to operate or handle the trap.

Failure to follow these safety precautions can result in serious injury.

Animal Handling Safety

- **Approach the trap cautiously** when an animal is captured, and wear protective gloves to prevent injury.
- **Avoid direct contact with animals**, especially if aggressive or frightened.

Safety Information, Restrictions, and Notices

Safety and Use Restrictions — DO NOT USE OCUTRAP:

- AS A SUBSTITUTE FOR LIFE SAFETY OR MEDICAL DEVICES.
- FOR ANY TYPE OF MEDICAL MONITORING OR LIFE-SUSTAINING APPLICATION.
- IN ANY WAY THAT VIOLATES FEDERAL, STATE, LOCAL, OR ADMINISTRATIVE LAWS, REGULATIONS, OR ORDINANCES. This includes, but is not limited to, laws governing wildlife and animal welfare, health and safety, data privacy, and security.
- IN A MANNER THAT INVOLVES CRIMINAL OR ILLEGAL ACTIVITIES.

OcuTrap is not responsible for injury during use.

DO NOT ATTACH OCUTRAP TO OR PLACE IT NEAR LIFE SAFETY DEVICES, MEDICAL MONITORING EQUIPMENT, OR OTHER SENSITIVE DEVICES where it could interfere with the operation, safety, or functionality of such equipment.

OcuTrap R2 Hardware Warranty

1. Warranty Coverage

OcuTrap, Inc. ("Company") warrants that the "OcuTrap R2" hardware will be free from defects in materials and workmanship under normal use and conditions for a period of 12 months from the date of original purchase by the end user ("Warranty Period").

2. Warranty Conditions

During the Warranty Period, the Company will repair or replace, at its discretion, any defective components or product covered by this warranty. Replacement parts or products may be new or reconditioned at the Company's discretion and will be warranted for the remainder of the original Warranty Period or 90 days from the date of repair or replacement, whichever is longer.

3. Limitations and Exclusions

This warranty does not cover the following:

- **Improper Installation:** damage caused by incorrect installation not in accordance with the Company's installation guidelines.
- **Misuse or Neglect:** issues arising from misuse, neglect, or improper maintenance.
- **Third-party Modifications:** defects caused by unauthorized modifications, adjustments, or repairs by the user or a third party.
- **Environmental Factors:** damage due to power failures, power surges, extreme weather conditions, or other environmental factors beyond the Company's control.
- **Normal Wear and Tear:** cosmetic damage or normal wear and tear not affecting the functionality of the OcuTrap R2.
- **Non-OcuTrap Parts or Accessories:** damage caused by the use of parts or accessories not authorized by OcuTrap, Inc.

4. Warranty Claims Process

To make a claim under this warranty, the customer must:

1. Contact OcuTrap Support at support@ocutrap.com with proof of purchase.
2. Provide a description of the issue, including any relevant photos or details to help diagnose the problem.
3. Ship the product to an authorized repair facility or return address if requested by the Company, with shipping label provided by OcuTrap.

5. Liability Limitation

The Company's liability under this warranty is limited solely to the repair or replacement of the defective product or parts at its discretion. The Company is not liable for any indirect, incidental, or consequential damages arising from the use of, or inability to use, the OcuTrap R2, including any loss of profits, business interruption, or damage to other property.

6. Governing Law

This warranty is governed by the laws of Texas without regard to its conflict of laws principles. Any disputes arising from this warranty will be subject to the exclusive jurisdiction of the courts located in the state of Texas.

7. Warranty Modifications

OcuTrap, Inc. reserves the right to modify the terms and conditions of this warranty at any time, with any such changes not applying retroactively to products purchased before the modification date.

8. Contact Information

For any questions or concerns regarding this warranty, please contact: **support@ocutrap.com**

FCC Compliance and Laser Safety

FCC Compliance Statement

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

- This device may not cause harmful interference.
- This device must accept any interference received, including interference that may cause undesired operation.

FCC ID

The FCC Identifier for this device is: **FCC ID: 2AEMI-B404X**

Modifications Warning

Changes or modifications not expressly approved by Particle Industries, Inc. could void the user's authority to operate the equipment.

RF Exposure Information

This equipment complies with FCC radiation exposure limits set forth for an uncontrolled environment. To maintain compliance, this device should be installed and operated with a minimum distance of 20 centimeters between the radiator and your body.

Antenna Information

The B404X SoM is designed to be used with external antennas. Ensure that the antennas used are of the same type and have equal or lesser gain than those tested to comply with FCC regulations.

Responsible Party Information

For compliance inquiries, please contact: Particle Industries, Inc., 325 9th St, San Francisco, CA 94103 USA

Additional Resources

For more detailed information, including test reports and certifications, please refer to the official FCC documentation for this device: <https://fccid.io/2AEMI-B404X>

Laser Safety Information

This product contains a laser emitter and corresponding drive circuitry. The laser output is designed to meet Class 1 laser safety limits under all reasonably foreseeable conditions, including single faults, in compliance with **IEC 60825-1:2014**.



Important Safety Instructions

- **Do not increase the laser output power by any means.**
- **Do not use any optics to focus the laser beam.**
- **Caution:** use of controls or adjustments, or performance of procedures other than those specified herein, may result in hazardous radiation exposure.

Compliance Information

This product complies with the following standards:

- **IEC 60825-1:2014**
- **21 CFR 1040.10 and 1040.11**, except for conformance with IEC 60825-1:2014 as described in the Laser Notice Number 56, dated May 8, 2019.
- **EN 60825-1:2014**, including **EN 60825-1:2014/A11:2021**
- **This product is not intended for children.**

OcuTrap relies on a third-party wireless service using radio technology, which is subject to transmission and service area limitations. This may include **interruptions or dropped connections** due to atmospheric, topographical, or environmental factors, cell site availability, cellular network equipment or installation, government regulations, system limitations, maintenance, or other conditions affecting wireless service functionality. Wireless service and features may not be available in all areas.



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